

Taiabur Rahman

AI in Healthcare | Medical Image | Fullstack Developer
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SUMMARY

AI-driven 3D medical imaging processing classification, segmentation, registration, cloud deployment (AWS, Docker), and cross-functional team collaboration in healthcare environment with Fullstack developer with 4+ years of experience in healthcare, SaaS platforms and DevOps.

EDUCATIONS

Erasmus Mundus Joint Master Degree in Medical Imaging and Applications (MAIA) Sep 2022 -Aug 2024

The University of Burgundy (France), the University of Cassino and Southern Lazio (Italy), and the University of Girona (Spain).

Field of Study: Medical Sensors, Machine and Deep Learning, Advance Image Analysis, Medical Image Segmentation and Registration, Computer Aided Diagnosis, Computer Aided Surgery and Medical Robotics, E-Health.

MSc. Software Engineering (SE)

Tianjin University, School of Computer Software, China

BSc. Computer Science & Engineering (CSE)

Bangladesh University (BU), Bangladesh

WORKSHOPS & SHORT PROGRAMS

17th EXCITE Summer School on Biomedical Imaging Sep 2023
Swiss Federal Institute of Technology Zurich (ETH Zurich)
University of Zurich, Switzerland

Winter School 2023 Feb 2023
University of Cassino and Southern Lazio-Italy

EMPLOYMENT HISTORY

Machine Learning Engineer Sep 2025
BLACKBIRD.AI, New York, United States · Remote
- Contributing to machine learning model development, data analysis, and continuous improvement of AI systems.

Graduate Research Intern Feb 2024
EQUIPE EMERGENTE NEUROGEMM INSERM UMR 1231, Dijon, France
Title: Deep Learning-Driven Automated Segmentation of High-Resolution 3D Histological Mouse Brain Volumes.

- Developed deep learning models for automated segmentation of 3D histological mouse brain volume
- Preprocessed large-scale high-resolution 3D image datasets
- Performed model training, tuning, and evaluation
- Collaborated with neuroscientists to validate biological accuracy
- Contributed to research documentation and presentations

Full Stack Developer (Full time) Dec 2017 - Aug 2022
Vision Eye Hospital, Bangladesh
- Developed and maintained Hospital Information Management System (HIMS) ERP solution in multiple branches .
- Built modules for prescription, diagnostics, surgical workflow, and patient record keeping.

RESEARCH PROJECTS

- * **Deep Learning-Driven Segmentation of Mouse Brain Volumes:** Developed a NNU-Net-based automated segmentation pipeline for high-resolution histological 3D mouse brain data using Python, PyTorch.
- * **Elastix-Based CT Image Registration:** Implemented a rigid and affine registration framework using SimpleITK and Elastix. [\[github\]](#)
- * **Human Brain Tissue Segmentation (3D volume MRI):** Built U-Net models using TensorFlow and PyTorch for white matter, gray matter, and CSF tissue segmentation. [\[github\]](#)
- * **Skin Lesion Classification (2d image):** Applied transfer learning techniques with VGG16 and ResNet50 using TensorFlow/ Keras to classify dermoscopic images. Used cross-validation and fine-tuning for performance improvement. [\[github\]](#)
- * **Retinal Image Analysis (2D image):** Developed a traditional computer vision pipeline using Python (thresholding, morphological operations) for segmented hard exudates. [\[github\]](#)
- * **Alzheimer's Disease Classification:** Combined MRI-based image features with gene expression data. Applied support vector machines (SVM) and ensemble techniques using Scikit-learn and Python. [\[github\]](#)
- * **Regression Model Exploration:** Built and compared linear regression and k-NN models using Scikit-learn for healthcare cost prediction, exploring evaluation metrics like MAE and RMSE. [\[github\]](#)
- * **Deep learning base Image Segmentation Review:** Reviewed traditional vs deep learning segmentation literature. Summarized techniques including region growing, active contours, and CNN-based U-Net architectures.

KEYWORD FOR ATS

AI, 3D Medical Imaging, 3D data analysis, Classification, segmentation, registration, PyTorch, Deep Learning, Python, React, VUE Cli, Flask, FastAPI, MySQL, Docker, CI/CD, REST APIs, SaaS, AWS

SEEKING OPPORTUNITIES

- R&D and innovation in Medical & healthcare tech (AI, imaging, software)
- Full Stack Software Engineering
- DevOps (CI/CD, Docker, AWS, automation)

TECHNICAL SKILLS

AI/LLM: Gemini, OpenAI

ML/DL: PyTorch, TensorFlow, Keras

Backend: Python (Flask, FastAPI, Django), PHP (Laravel)

Frontend: Vue.js, Typescript, JavaScript, jQuery, React

DevOps & Cloud: AWS Docker, CI/CD, Linux

Database: MySQL, PostgreSQL, SQLite

Tools: Git, REST API, Nginx, Apache

LANGUAGE

- Bengali – Native
- English – Intermediate (B2)
- French – Beginner (A1 – currently improving)

SOFTWARE PROJECTS

AI-Powered Platform for Medical Image Quality Control & Analysis
<https://ai-miga.com>

Enhancing diagnostic accuracy and research reliability with intelligent, automated, and secure imaging solutions.

Hospital Information Management System (Healthcare Data)

A web-based platform actively used in 5 hospitals across Bangladesh. Tech Stack: Laravel, Vue.js, MySQL

LINKS

- + GitHub: <https://github.com/taiaurbd>
- + LinkedIn: <https://www.linkedin.com/in/taiaur-rahman>
- + Portfolio: <https://taiaurbd.github.io>